

TITLE PAGE

- Problem Statement ID – SIH25022
- Problem Statement Title – Maximizing Section Throughput using AI-Powered Precise Train Traffic Control (Railways)
- Theme – Transportation
- PS Category – Software
- Team ID – 74516
- Team Name (Registered on portal) – SignalX_1





IDEA TITLE:

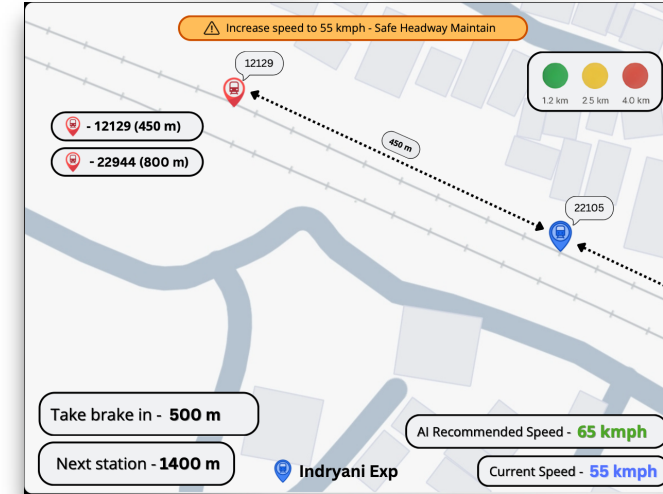
Preventive Rail Traffic Control (PRTC)



Proposed Solution

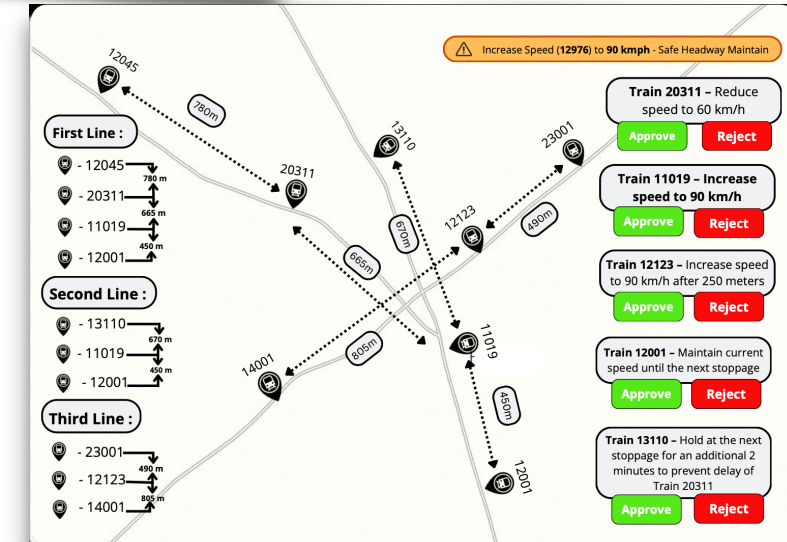
- Presently two systems are used:
 - Absolute Block System** → Only one train between two stations.
 - Automatic Block System** → Multiple trains allowed but follow fixed block signals.
- Problem: Section capacity remains underutilized, trains wait despite available track length.
- Proposed AI-Powered Solution:**
 - Real-time tracking of trains (speed, position, braking distance, train length).
 - AI calculates safe speed and headway for each following train.
 - Multiple trains can run between two stations **safely and optimally**.
 - System gives live speed recommendations to operators/control panel.

🌟 *Transforms real operations from reactive to productive and preventive*



RAILMATE -
AI-powered smart assistant for locomotive pilots

RAILCOMMAND -
AI-powered traffic control display for controllers

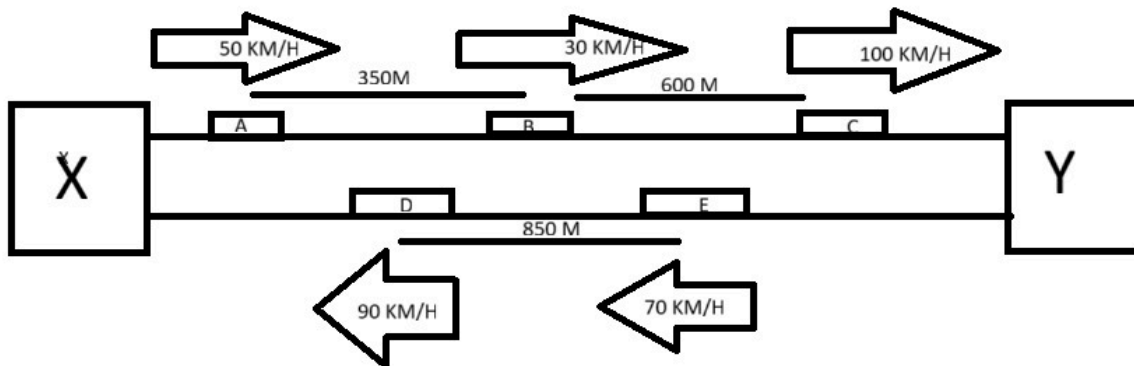


Real-Time Data Collection

- Each train fitted with onboard tablet/device.
- Captures speed, braking type, train length, acceleration, and reaction time.
- Data synced with central control room and loco pilot display.

Stopping Distance Calculation (Formulas Used):

1. Reaction Distance $\rightarrow S_{reaction} = u \times t_{reaction}$
 - $u = \frac{speed (km/h) \times 1000}{3600}$
2. Braking Distance $\rightarrow s = \frac{u^2}{2a}$
3. Total Stopping Distance $\rightarrow S_{total} = S_{reaction} + s$



AI-Powered Engine

- Calculates real-time **safe distance** using above formulas.
- Predicts **conflicts** and suggests preventive actions (speed adjustment / regulated halt).
- Considers braking differences: Air Brakes, Electro Pneumatic, Vacuum Brakes.
- Continuously improves accuracy using **train-specific learning**.

Technology Stack

- **Deep Reinforcement Learning (DRL):** Optimization of speed profiles.
- **Operations Research (OR) Models:** Constraint handling for headway & safety.
- **Digital Twins:** Virtual simulation of section for real-time decision testing.
- **APIs:** Integration with train GPS, signalling data, braking sensors.

Visualization & Control

- Loco pilot screen: AI recommended safe speed + headway info.
- Control room dashboard: Live monitoring of all trains + override option.
- Final authority: Loco pilot follows AI guidance, but station master/control can adjust.



FEASIBILITY AND VIABILITY



- **Feasibility**

- Uses existing **GPS modules, train tablets, and track circuits**
→ no need for heavy infrastructure change.
- Compatible with both **Absolute Block System** and **Automatic Block System**.
- AI engine deployable as a **software layer** over current signalling.

- **Pilot Deployment**

- Start on **busy corridors** (e.g., Delhi–Kanpur, Mumbai–Pune) with high congestion.
- Demonstrate safe simultaneous running of 3–5 trains in one section.

- **Challenges & Mitigation**

- **Data Availability:** Begin with simulation, then integrate live Indian Railways data gradually.
- **Safety-Critical Nature:** Conservative AI models, human approval before execution.
- **Scaling Across Zones:** Modular design → extend corridor by corridor.

- **Viability**

- Scalable to **nationwide deployment** after successful pilots.
- Long-term adoption aligns with Government’s **“Digital Railways” vision**.
- Low-cost enhancement compared to upgrading signalling hardware.

- **Key Features**

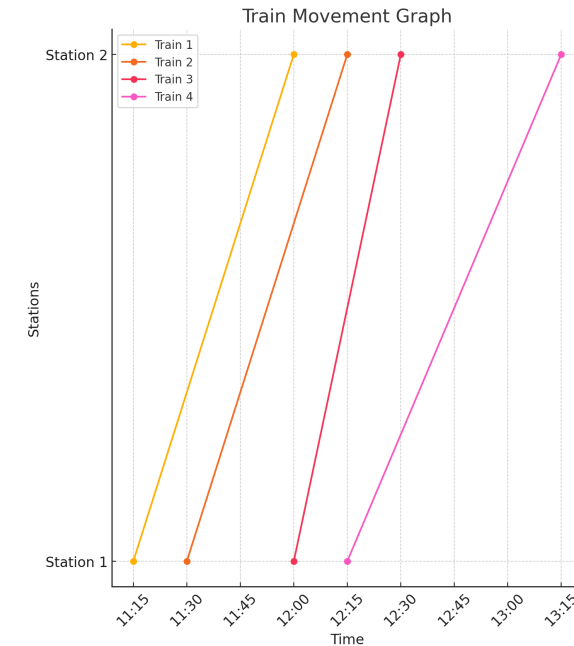
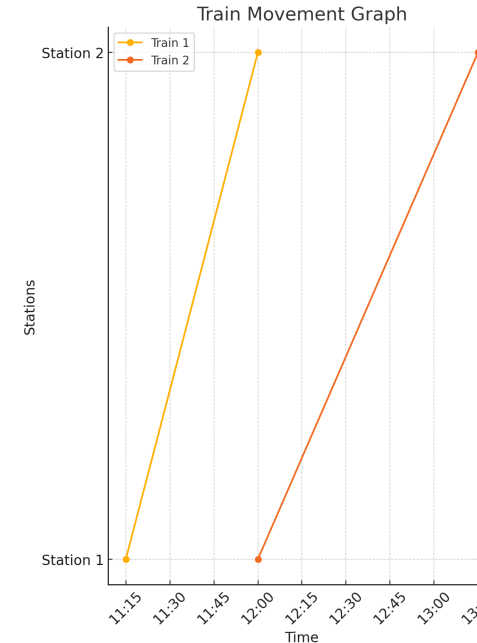
- Train-specific recommendations based on braking system (Air, Electro Pneumatic, Vacuum).
- **Human-in-loop** – operator confirms AI suggestions for fail-safe operation.



IMPACT AND BENEFITS



- **Operational Impact**
 - Increased **section throughput** → more trains can run on the same track safely.
 - Reduced **headway gaps** between trains without compromising safety.
 - Improved **punctuality and reliability** of passenger trains.
- **Economic Benefits**
 - Reduced delays → cost savings in fuel and crew hours.
 - Boosts Indian Railways' operational efficiency with minimal infrastructure cost.
- **Passenger & Social Benefits**
 - On-time arrivals → improved **passenger satisfaction and fewer cancellations**.
 - Enhanced trust in Indian Railways as a modern, efficient transport system.
- **Environmental Benefits**
 - Reduced fuel wastage from idling and repeated acceleration and lower **carbon emissions**
 - Supports India's **sustainable and green transportation goals**.



- **Strategic Benefits**
 - Step towards **AI-driven Digital Railways**.
 - Technology scalable for **nationwide adoption**.
 - Future-ready system integrating with **smart mobility and multimodal transport**.



RESEARCH AND REFERENCES



Topic : INDIAN RAILWAYS PERMANENT WAY MANUAL ✨

Link : <https://rdso.indianrailways.gov.in/works/uploads/File/p%29%20irpwm-2004.pdf>

Topic : BREAKING SYSTEM IN INDIAN RAILWAYS

Link : <https://www.ijert.org/research/braking-systems-in-railway-vehicles-IJERTV4IS010360.pdf?hl=en>

Topic : SIGNALLING IN INDIAN RAILWAYS

Link : <https://iriset.railnet.gov.in/content/coursmat/cli/coumate/CLI-01.pdf>

Topic : ABSOLUTE BLOCK SYSTEM

Link : https://indianrailways.gov.in/railwayboard/uploads/directorate/safety/CR/GR_SR_Chap-8_CR.pdf

Topic : AUTOMATIC BLOCK SYSTEM

Link : <https://ner.indianrailways.gov.in/uploads/files/1597319285976-Automatic%20Intermediate%20Block%20Signalling.pdf>

Topic : TRAIN WORKING SYSTEM

Link:- https://www.irfca.org/faq/faq-signal4_b.html